

div by 2 and 3 $\rightarrow 6, 12, 18, 24, \dots = A \cap B$

div by 2 or 3 $\rightarrow 2, 3, 4, 6, 8, 9, 10, 12, \dots = A \cup B$

$\cap$ $\rightarrow$ and $\rightarrow$ all conditions to be satisfied

$\cup$ $\rightarrow$ or $\rightarrow$ at least one of the condition is satisfied

formula on 2 sets



$$n(A-B) = n(A) - n(A \cap B)$$

$$n(B-A) = n(B) - n(A \cap B)$$



$$n(A \cup B) = n(A-B) + n(A \cap B) + n(B-A)$$

$$n(A \cup B) = n(A) + n(B) - n(A \cap B)$$

$$A \rightarrow \{2, 4, 6, 8, \dots\}$$

$$B \rightarrow \{3, 6, 9, 12, \dots\}$$



$$n(A^c \cap B^c) = n(U) - n(A \cup B)$$

$$n(A^c) = n(U) - n(A)$$

Q In a class of 50 students, 40 plays cricket, 30 plays football and 25 plays both. Find no of students that play

(i) football or cricket

(ii) only cricket

(iii) not play football

(iv) does not play anything

(v) plays only one sports.

$$n(C) = 40$$

$$n(F) = 30$$

$$n(C \cap F) = 25$$

$$n(U) = 50$$

$$n(C \cup F) = ?$$

$$n(C - F) = ?$$

$$n(F^c) = ?$$

$$n(C^c \cap F^c) = ?$$

$$n(C - F) + n(F - C) = ?$$

$$\begin{aligned} n(C \cup F) &= n(C) + n(F) - n(C \cap F) \\ &= 40 + 30 - 25 \\ &= 45 \end{aligned}$$

$$\begin{aligned} n(C - F) &= n(C) - n(C \cap F) \\ &= 40 - 25 \\ &= 15 \end{aligned}$$

$$\begin{aligned} n(F^c) &= n(U) - n(F) \\ &= 50 - 30 \\ &= 20 \end{aligned}$$

$$\begin{aligned} n(C^c \cap F^c) &= n(U) - n(C \cup F) \\ &= 50 - 45 \\ &= 5 \end{aligned}$$

$$\begin{aligned} n(C - F) + n(F - C) &= n(C) - n(C \cap F) \\ &\quad + n(F) - n(C \cap F) \\ &= 40 - 25 + 30 - 25 \\ &= 20 \end{aligned}$$

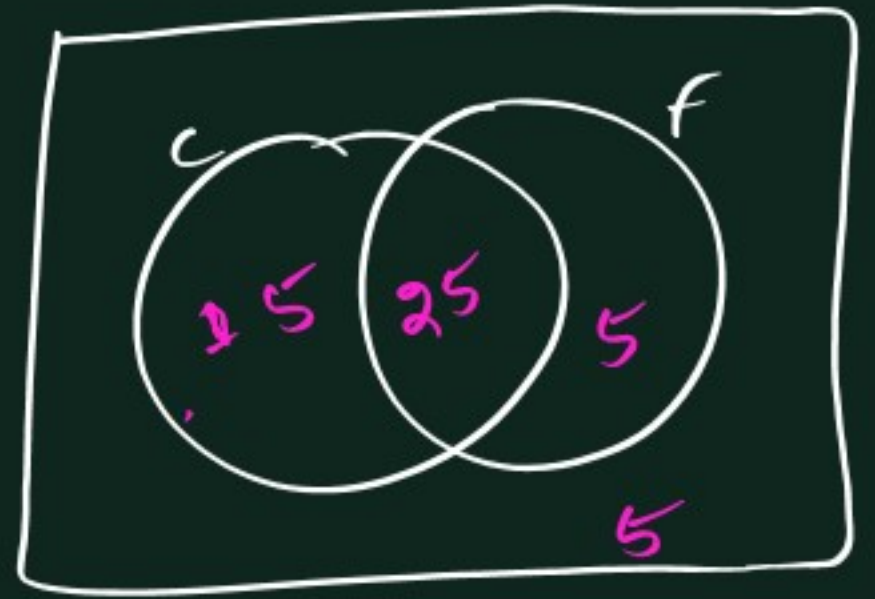
a) cricket or football = $15 + 25 + 5 = 45$

b) only cricket = 15

c) not play football = $15 + 5 = 20$

d) does not play anything = 5

e) plays one sports = $15 + 5 = 20$



Q 9 In a class of 50, everyone drink either tea or coffee. If 31 students drink coffee and 38 students drink tea. Then find no of students who drink

a) only coffee $\rightarrow 12$

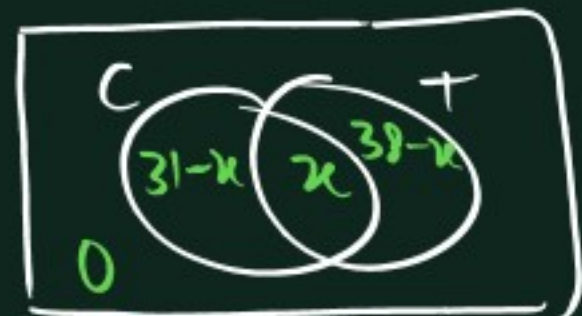
b) only one drink $\rightarrow 12 + 19 = 31$

c) drinks both $\rightarrow 19$

$$0 + (31 - x) + x + (38 - x) = 50$$

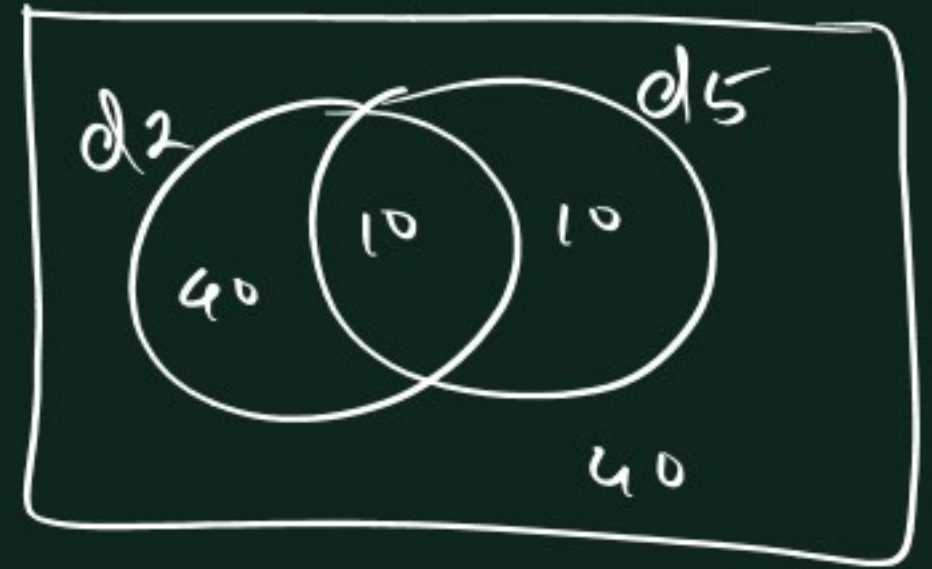
$$69 - x = 50$$

$$x = 19$$

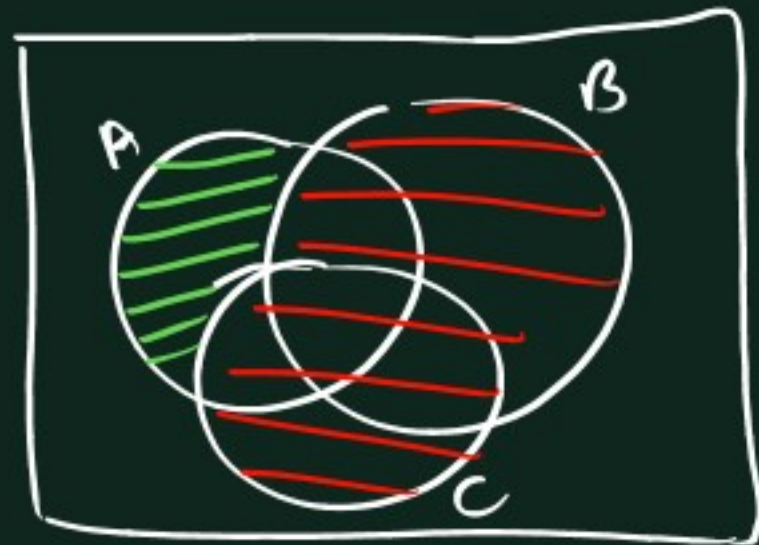


Q $U = \{1, 2, 3, 4, \dots, 100\}$, find no of numbers that are divisible by

- a) neither 2 nor 5 $\rightarrow 40$
b) div by 2 but not 5 $\rightarrow 40$



for 3 sets



Exactly one



Exactly two



Exactly 3

at least one spots
 $A \cup B \cup C$



$$n(A \cup B \cup C) = \text{only } A + n(B \cup C)$$

$$= n(A) - n(A \cap B) - n(A \cap C) + n(A \cap B \cap C) \\ + n(B) + n(C) - n(B \cap C)$$

$$n(A \cup B \cup C) = \underline{n(A) + n(B) + n(C)} - \underline{n(A \cap B) - n(B \cap C) - n(C \cap A) + n(A \cap B \cap C)}$$

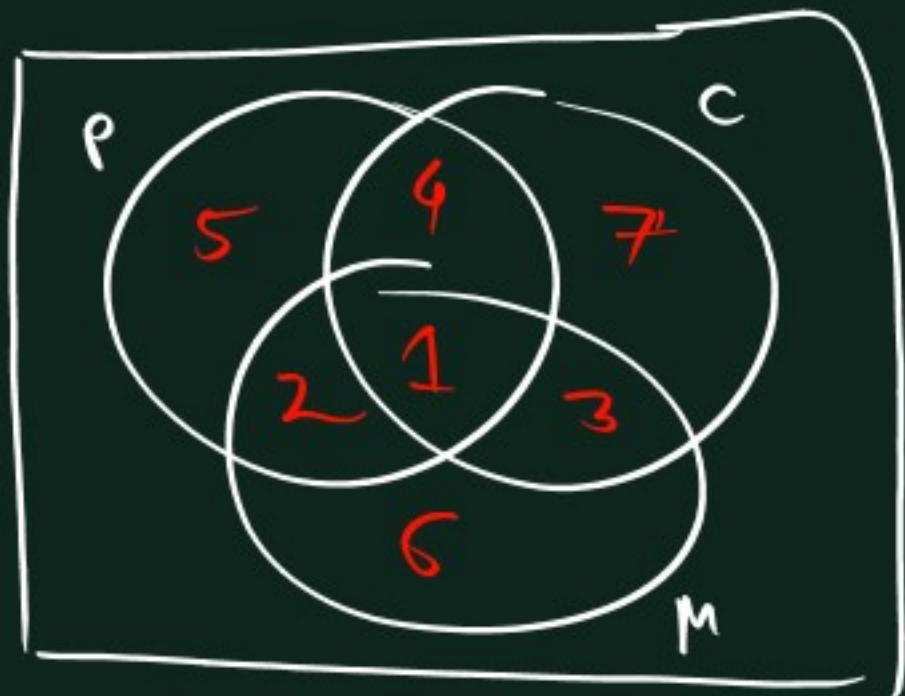
$$\text{only } A = n(A) - n(A \cap B) - n(A \cap C) + n(A \cap B \cap C)$$

In a school, a student can select maximum 3 elective subjects, that are maths, Physics and chemistry.

→ 12, 12 and 15 students chose M, P, C respectively

→ 3 took M and P, 5 took P and C, 4 took C and M and 1 selected all 3 subjects.

Find number of students that have selected



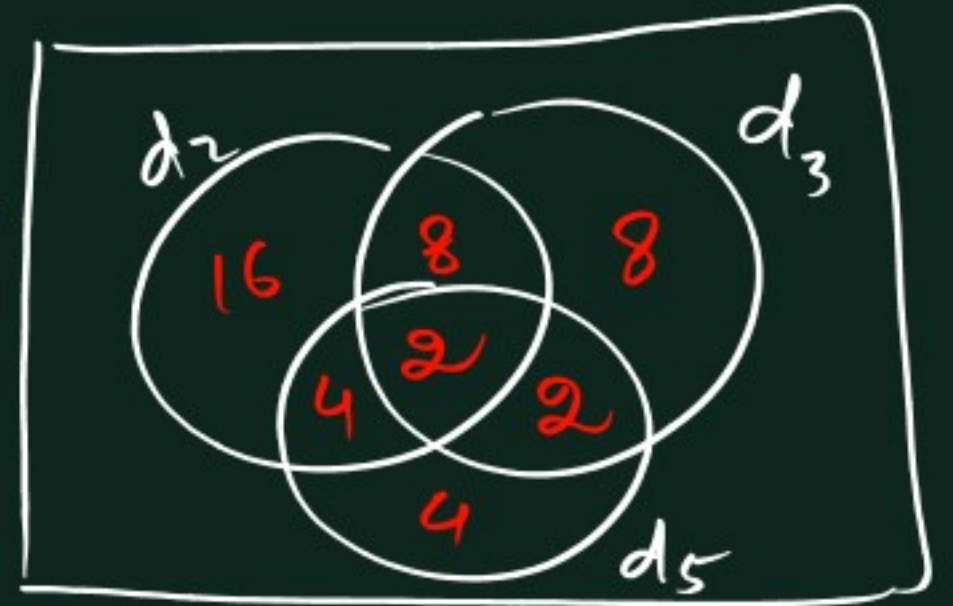
- a) P or C or M $\rightarrow 5 + 4 + 7 + 2 + 1 + 3 + 6 = 28$
- b) only maths = 6
- c) at least 2 subjects = $1 + 4 + 2 + 3 = 10$
- d) selected one subject = $5 + 6 + 7 = 18$

Q $U = \{1, 2, 3, \dots, 60\}$, Find no of numbers that are

a) div by 2 or 3 or 5 $\rightarrow 16 + 8 + 8 + 4 + 2 + 2 + 4 = 44$

b) div by 3 or 5 but not div by 2 $= 8 + 2 + 4 = 14$

c) div by 2 and 3 but not div by 5 $= 8$

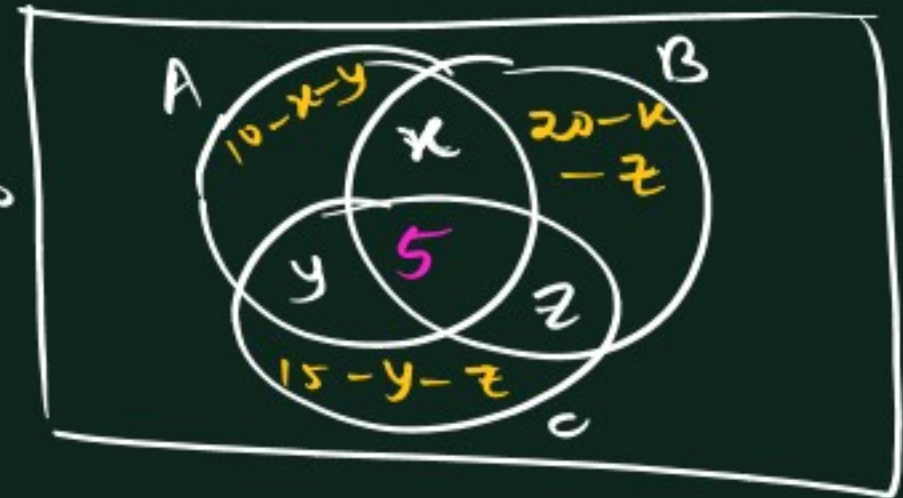


Q In a village 15, 25 and 20 people reads the newspaper A, B, C respectively and 5 person reads all 3 newspaper. If the total number of person reading newspaper is 40. Then find number of people who reads exactly 2 newspaper.

$$5 + \cancel{x} + \cancel{y} + \cancel{z} + (10 - \cancel{x} - \cancel{y}) + (15 - \cancel{y} - \cancel{z}) + (20 - \cancel{x} - \cancel{z}) = 40$$

$$\Rightarrow 5 + 10 + 15 + 20 - 40 - y - x - z = 0$$

$$\Rightarrow 10 = x + y + z$$

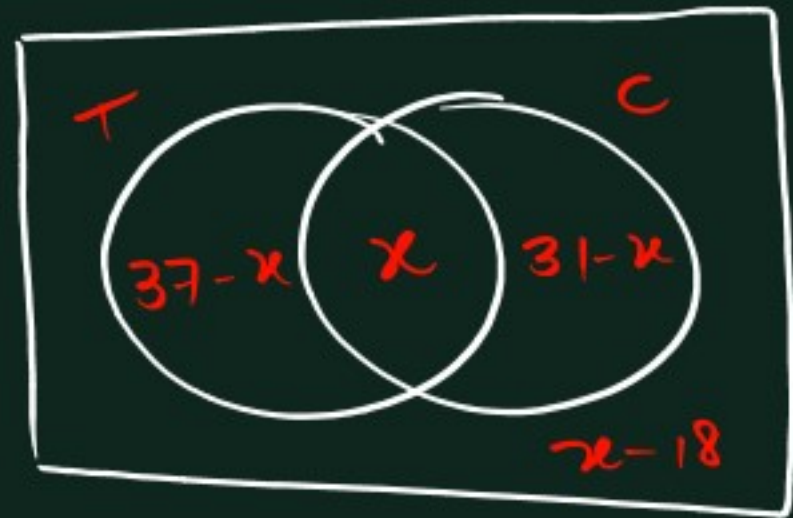


$$\begin{aligned} \text{Exactly 2 newspaper} &= x + y + z \\ &= \underline{10} \end{aligned}$$

Q There are 50 students in a class and 31 of them drink coffee and 37 of them drink tea. let 'x' be the number of people drinking tea and coffee both. Find minimum and maximum value of 'x'.

max value is 31

min value is 18



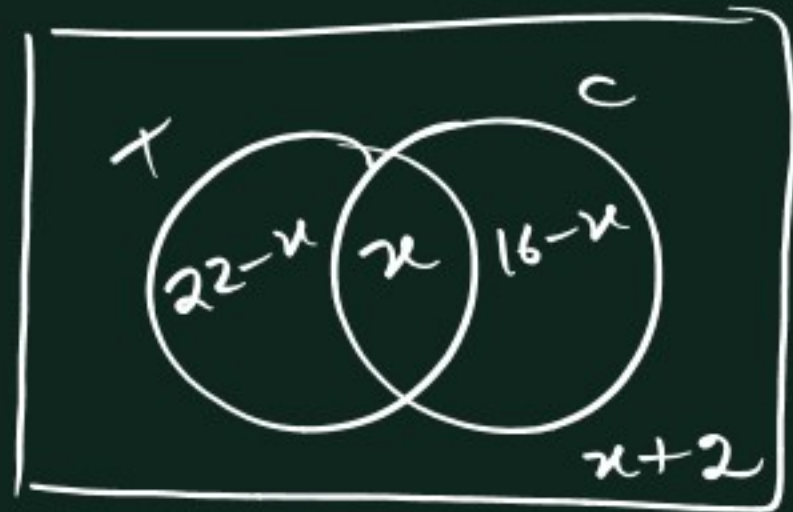
$$\begin{aligned} y &= 50 - (37-x) - x - (31-x) \\ &= 50 - 37 + \cancel{x} - \cancel{x} - 31 + x \\ &= x - 18 \end{aligned}$$

40 students
 ↙ ↘
 16 Coffee 22 Tea

$$\max(x) = 16$$

$$\min(x) = 0$$

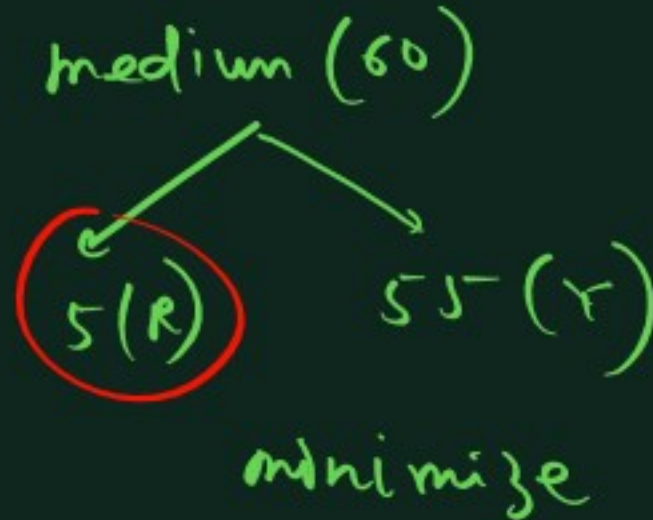
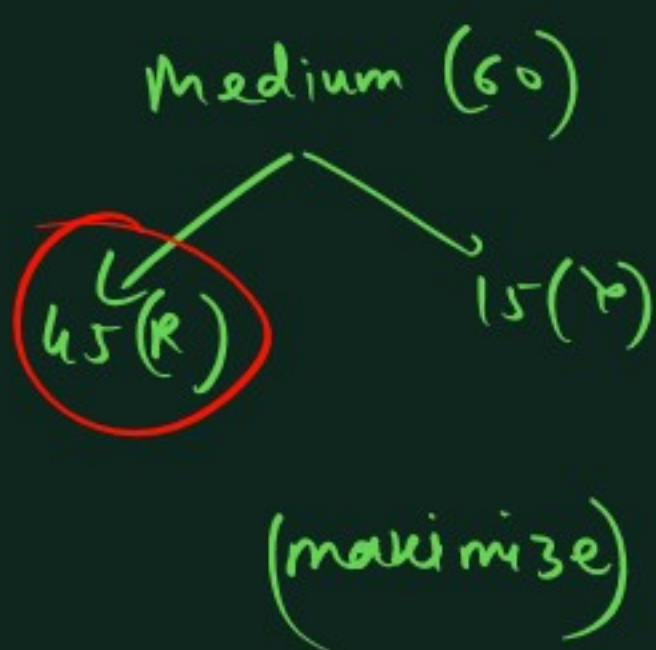
both = x , min and max of x .



Q T shirt (Total = 100)

M1 Red → 45 | L → 40
 Yellow → 55 | M → 60

$\max(\text{Red}, M) = ?$
 $\min(\text{Red}, M) = ?$



M2

$$y = 40 - (22 - x) - x - (16 - x)$$

$$= 40 - 22 + x - x - 16 + x$$

$$= x + 2$$

	Red	Yellow
M	x	$60 - x$
L	$45 - x$	$x - 5$

$$y + (60 - x) = 55$$

$$y = x - 5$$

$$\max = 45$$

$$\min = 5$$

Sequence and series

Sequence: ordered list of number.

eg: 1, 3, 5, 7, 9, ...

1, 15, 2, 7, -5, 6, ...

Progression: Sequence with pattern is progression

eg: 1, 3, 5, 7, 9, (11) - -

0, 1, 1, 2, 3, 5, 8, 13, 21, (34) -
↑

Series: → addition of terms in progression is series

eg: 2 + 4 + 8 + 16 + 32 + ...

eg: 1 - 3 + 5 - 7 + 9 - 11 + ...

Q Write first 5 terms

(i) $a_n = \frac{1}{2^n + 1}$

(ii) $a_{n+1} = a_n + n$ for $n \geq 1$
and $a_1 = 5$

(i) soln

$$a_1 = \frac{1}{2^1 + 1} = \frac{1}{3}$$

$$a_2 = \frac{1}{2^2 + 1} = \frac{1}{5}$$

$$a_3 = \frac{1}{2^3 + 1} = \frac{1}{9}$$

$$a_4 = \frac{1}{2^4 + 1} = \frac{1}{17}$$

$$a_5 = \frac{1}{2^5 + 1} = \frac{1}{33}$$

$$a_{n+1} = a_n + n, \quad n \geq 1, \quad a_1 = 5$$

$$n=1, \quad a_2 = a_1 + 1 = 5 + 1 = 6$$

$$n=2, \quad a_3 = a_2 + 2 = 6 + 2 = 8$$

$$n=3, \quad a_4 = a_3 + 3 = 8 + 3 = 11$$

$$n=4, \quad a_5 = a_4 + 4 = 11 + 4 = 15$$

S_n = sum of first n terms

$$S_n = \left[t_1 + t_2 + t_3 + \dots + t_{n-1} \right] + t_n$$

$$= S_{n-1} + t_n$$

$$t_n = S_n - S_{n-1} \rightarrow \text{valid for all.}$$

§ $S_n = n^2 + 2n$, then find t_{10}
and also t_n .

$$S_n = n^2 + 2n$$

$$T_n = S_n - S_{n-1}$$

$$n=10, T_{10} = S_{10} - S_9$$

$$= 120 - 99$$

$$= 21$$

$$S_{10} = 10^2 + 2 \times 10 = 120$$

$$S_9 = 9^2 + 2 \times 9 = 99$$

$$T_n = S_n - S_{n-1}$$

$$= (n^2 + 2n) - (n^2 - 1)$$

$$= \cancel{n^2} + 2n - \cancel{n^2} + 1$$

$$= 2n + 1$$

$$S_n = n^2 + 2n$$

$$S_{n-1} = (n-1)^2 + 2(n-1)$$

$$= \cancel{n^2} - \cancel{2n} + 1 + \cancel{2n} - 2$$

$$= n^2 - 1$$

AP:

$a, a+d, a+2d, a+3d, \dots, a+(n-1)d$

$t_n = a + (n-1)d$, $t_n \rightarrow$ is linear in AP

$$S_n = a + (a+d) + (a+2d) + \dots + \{a+(n-1)d\}$$

$$S_n = \{a+(n-1)d\} + \{a+(n-2)d\} + \{a+(n-3)d\} + \dots + a$$

$$2S_n = \{2a+(n-1)d\} + \{2a+(n-1)d\} + \{2a+(n-1)d\} + \dots + \{2a+(n-1)d\}$$

$$2S_n = n \{2a+(n-1)d\}$$

$$\frac{n}{2} \{2a+(n-1)d\}$$

